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BONDING TOOL FOR BONDING DIES ONTO SUBSTRATES AND METHODS OF USE

Abstract

Embodiments of the present disclosure provide a bonding tool for bonding dies onto substrates and a method that includes positioning a substrate and a plurality of dies over a substrate support of a bonding tool, wherein the plurality of dies are disposed on the substrate, and the substrate is surrounded by an edge ring, inflating a membrane of the bonding tool disposed above the substrate and the edge ring to expand the membrane downwards, wherein a polyetheretherketone (PEEK) layer attached to the membrane rests on the edge ring, and applying a force from the inflating membrane on the plurality of dies via the PEEK layer.

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Background/Summary

BACKGROUND

[0001] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

[0002] Therefore, there is a need to improve processing and manufacturing ICs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIGS. 1A-1E illustrate views of various stages of bonding dies onto a substrate, in accordance with some embodiments.

[0005] FIG. 2 is a flowchart that illustrates a method of bonding dies onto a substrate, in accordance with some embodiments.

[0006] FIG. 3 is a perspective view of a membrane assembly that includes a membrane with a polymer layer attached, in accordance with some embodiments.

[0007] FIG. 4 is a partial sectional view that illustrates an inflation state of bonding a die onto the substrate, in accordance with some embodiments.

[0008] FIG. 5 is a schematic partial sectional view that illustrates a force application state of bonding a die onto the substrate, in accordance with some embodiments.

[0009] FIGS. 6A-6E illustrate views of various stages of bonding dies onto a substrate, in accordance with some embodiments.

[0010] FIG. 7 is a flowchart that illustrates an alternative method of bonding dies onto a substrate, in accordance with some embodiments.

[0011] FIG. 8 is a schematic sectional view that illustrates a first bonding step of forming a semiconductor device package, in accordance with some embodiments.

[0012] FIG. 9 is a schematic sectional view that illustrates a second bonding step of forming the semiconductor device package, in accordance with some embodiments.

DETAILED DESCRIPTION

[0013] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of

simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0014] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “over,” “on,” “top,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0015] Embodiments of this disclosure provide a bonding tool for bonding dies onto substrates and associated methods. According to the methods described herein, particularly the use of a polymer layer between a bonding tool membrane and a die that is to be bonded, substantial increases in die yield, determined through wafer assurance test (WAT), can be achieved. For example, testing has shown increases in die yield of 10%-20% and die yields of 95%-100%. In addition, the methods described herein can reduce die shift and enhance final bonding strength on the order of about 0.5 J/m.^{sup.2}.

[0016] FIGS. **1A-1E** illustrate views of various stages of bonding dies onto a substrate, in accordance with some embodiments. FIG. **2** is a flowchart that illustrates a method **200** of bonding dies onto a substrate, in accordance with some embodiments. Referring to FIG. **1A** and FIG. **2** (block **210**), a substrate and a plurality of dies that are to be bonded to the substrate are transferred, via a robot handler, from a factory interface to a substrate support of a bonding tool. As shown in FIG. **1A**, a bonding tool **100** (e.g., a die press) includes a lower tray **102**, an upper tray **104**, and a seal ring **106** between the lower tray **102** and the upper tray **104**. The lower tray **102** and/or the upper tray **104** are able to be moved away from each other and towards each other to open and close the bonding tool **100**, respectively. For example, the lower tray **102** may be moved downwards in a substantially vertical direction and/or the upper tray **104** may be moved upwards in a substantially opposite vertical direction. The seal ring **106** is configured to form an airtight seal and/or a fluid tight seal between the lower tray **102** and the upper tray **104** when the lower tray **102** and the upper tray **104** are moved to a closed position of the bonding tool **100**. The closed position is shown in FIGS. **1A-1E**. The seal ring **106** may be an annular, elastomeric seal (e.g., an o-ring seal). A processing region **108** is defined between and/or within the lower tray **102** and the upper tray **104** and sealed by the seal ring **106**. The processing region **108** may be in fluid communication with processing equipment attached to the bonding tool **100** (e.g., a pump, vacuum pump, inert gas source, valves, sensors, and/or other equipment).

[0017] The lower tray **102** includes a substrate support **110** attached to the lower tray **102** and an edge ring **112** on the substrate support **110**. In some embodiments, the substrate support **110** and the edge ring **112** may be separate pieces that are attached together. For example, the edge ring **112** may be attached to an upper surface of the substrate support **110** that is facing away from the lower tray **102**. In some other embodiments, the substrate support **110** and the edge ring **112** may be a single, unitary piece. The upper tray **104** includes a membrane **114** (which also may be referred to herein as a “cushion”) attached to the upper tray **104** and a polymer layer **116** attached to the membrane **114**. The membrane **114** is disposed above the substrate support **110** and the edge ring **112**. For example, the membrane **114** may be attached to a lower surface of the upper tray **104** that is facing towards the lower tray **102**. Likewise, the polymer layer **116** may be attached to a lower surface of the membrane **114**. The process described in connection with FIGS. **1A-1E** and FIG. **2** may be referred to as an “auto handling” process because the polymer layer **116** is fixedly attached to the membrane **114** and is not handled by an operator during the process. The membrane **114** may be formed of a material that has relatively high elasticity compared to the polymer layer **116**. For example, the membrane **114** may include primarily silicone. The polymer layer **116** may be formed of a material that is different from the material of the membrane **114**. For example, the polymer

layer **116** may have an elasticity substantially smaller than the elasticity of the membrane **114** (e.g., a ratio of elasticity between the membrane **114** and the polymer layer **116** may be greater than $10\times$, such as within a range between about $10\times$ and about $100\times$). The polymer layer **116** may include primarily polyetheretherketone (PEEK). Based on the difference in elasticity between the membrane **114** and the polymer layer **116**, the membrane **114** is configured to expand, in volume, to a far greater extent than the polymer layer **116**. To account for this differential, the polymer layer **116** may be attached to the membrane **114** with a bonding technique that is able to withstand the relative difference in expansion between the two materials.

[0018] A substrate **118** and a plurality of dies **120** are positioned over the substrate support **110**. The plurality of dies **120** are disposed on the substrate **118**. For example, the plurality of dies **120** may be positioned on a top surface of the substrate **118** that is facing away from the substrate support **110**. The substrate **118** is surrounded by the edge ring **112**. The membrane **114** is disposed above the substrate **118** and the edge ring **112**. As shown in FIG. **1A**, the edge ring **112** extends above the plurality of dies **120**. For example, as described in more detail below, the edge ring **112** has a height substantially greater than a combined height of the substrate **118** and a die of the plurality of dies **120**.

[0019] The substrate **118** may include a semiconductor substrate, such as a wafer, chip, system on chip (SoC), or integrated system on chip (SoIC), among other examples. The plurality of dies **120** may include a semiconductor die that is to be attached to the semiconductor substrate, such as a wafer, chip, SoC, or SoIC, among other examples. The plurality of dies **120** may be arranged with respect to the substrate **118** according to a device layout that is to be formed using the die bonding process. For example, the plurality of dies **120** may be spaced apart across a diameter of the substrate **118**.

[0020] The lower tray **102** and the upper tray **104** may include respective heating elements that enable respective temperatures of the lower tray **102** and the upper tray **104** to be controlled independently. In some embodiments, die bonding using the bonding tool **100** described herein includes a low temperature anneal. For example, the respective temperatures of the lower tray **102** and the upper tray **104** may be within a range between about 130°C. and about 150°C. , such as about 140°C. In some embodiments, the respective temperatures of the lower tray **102** and the upper tray **104** may be set as the processing temperature for bonding the plurality of dies **120** to the substrate **118**. In some embodiments, the glass transition temperature (T_c) of the material of the polymer layer **116** is within a range between about -40°C. and about $+30^{\circ}\text{C.}$ with respect to T_c , such as within a range between about -30°C. and about $+20^{\circ}\text{C.}$ with respect to T_c , such as within a range between about -20°C. and about $+10^{\circ}\text{C.}$ with respect to T_c , such as within a range between about -10°C. and about $+10^{\circ}\text{C.}$ with respect to T_c . Thus, in some embodiments, the material of the polymer layer **116** is determined by the processing temperature of the die bonding process. If the T_c of the material of the polymer layer **116** is substantially greater than the processing temperature, such as about 50°C. greater than the processing temperature, the polymer layer **116** may break under a force F exerted by the membrane **114**. On the other hand, if the T_c of the material of the polymer layer **116** is substantially less than the processing temperature, such as about 50°C. less than the processing temperature, the polymer layer **116** may become too soft to transfer the force F uniformly to the plurality of dies **120**.

[0021] As shown in FIG. **1A**, a release film **122** is positioned (e.g., suspended) between the polymer layer **116** and the plurality of dies **120**. The release film **122** may be formed of a material that does not adhere to the polymer layer **116**, the substrate **118**, or the plurality of dies **120**. For example, without the release film **122**, the polymer layer **116** may be susceptible to adhering to and causing damage to the substrate **118** and/or the plurality of dies **120** during the die bonding process. The release film **122** may be a plastic material that is different from the polymer layer **116**. For example, the release film **122** may include primarily polytetrafluoroethylene (PTFE). In some embodiments, the release film **122** is a low-friction material (e.g., characterized with a smooth

and/or slippery texture). In some embodiments, T_c of the release film **122** may be within a range between about $2\times$ and about $5\times$ compared to T_c of the polymer layer **116**. The release film **122** may be one time use only and is configured to be replaced after each bonding step. For example, a roll of the release film **122** may be located at one end of the processing region **108**, and the release film **122** is pulled across the processing region **108** to an opposite end of the processing region **108**. The release film **122** may be pulled by any device, such as a roller. In some embodiments, after each die bonding process, the used release film **122** is pulled away from the processing region **108**, and a new release film **122** from the roll is positioned above the substrate support **110**.

[0022] Referring to FIG. **1B** and FIG. **2** (block **220**), the membrane **114** is inflated to expand the membrane **114**, and the polymer layer **116** attached to the membrane **114**, downwards towards the substrate **118**. As the membrane **114** is inflated, a vertical distance between a bottom surface of the polymer layer **116** and a top surface of the edge ring **112** decreases. During inflation of the membrane **114**, the polymer layer **116** contacts the release film **122**. After contacting the polymer layer **116**, the release film **122** moves downwards, with the polymer layer **116**, towards the substrate **118**. When the membrane **114** is inflated to a threshold volume (or threshold pressure), the polymer layer **116** contacts the edge ring **112** (e.g., via the release film **122**), as shown in FIG. **1B**. At this stage, the polymer layer **116** rests on the edge ring **112**, and the release film **122** is disposed between the polymer layer **116** and the edge ring **112**. For example, a diameter of the polymer layer **116** may be greater than an inner diameter of the edge ring **112** such that an outer portion of the polymer layer **116** overlaps with and rests on an inner portion of the edge ring **112**. In some embodiments, the membrane **114** may be inflated by providing pressurized gas (e.g., inert gas or air) inside the membrane **114**. For example, inflation of the membrane **114** may be based on a pressure differential between the membrane **114** and the processing region **108**. The upper tray **104** of the bonding tool **100** may include one or more gas connections that are in fluid communication between the inside of the membrane **114** and inflation equipment attached to the bonding tool **100** (e.g., a pump, vacuum pump, pressurized gas source, valves, sensors, and/or other equipment).

[0023] When the polymer layer **116** first contacts the edge ring **112** via the release film **122**, as shown in FIG. **1B**, there is a gap between a top surface of a die, of the plurality of dies **120**, and the bottom surface of the release film **122**. In some embodiments, the gap may range from about $80\text{ }\mu\text{m}$ to about $120\text{ }\mu\text{m}$. The membrane **114** may continue to be inflated after reaching the edge ring **112** causing an inner radial portion of the polymer layer **116** to advance across (or close) the gap to rest on the plurality of dies **120** and cause a force to be applied on the plurality of dies **120** as described in more detail below.

[0024] In some embodiments, inflating the membrane **114** at block **220** may include a first vacuum process that reduces pressure in the processing region **108** from atmospheric pressure (760 Torr) to about 5000 mTorr and a second vacuum process that further reduces pressure to less than about 1000 mTorr, such as less than about 100 mTorr, such as within a range between about 10 mTorr and about 100 mTorr. As the pressure in the processing region **108** is being reduced, the membrane **114** may be inflated simultaneously. In some embodiments, the vacuum and inflation process described above may occur over a total time period within a range between about 2 minutes and about 6 minutes, such as between about 3 minutes and about 5 minutes, such as about 4 minutes. In some embodiments, the first vacuum process may occur over a first time period T_1 within a range between about 30 seconds and about 90 seconds, such as about 60 seconds. In some embodiments, the second vacuum process and inflation process, which can take place simultaneously, may occur over a second time period T_2 within a range between about 2 minutes and about 4 minutes, such as about 3 minutes.

[0025] Referring to FIG. **1C** and FIG. **2** (block **230**), a force F is applied from the inflating membrane **114** on the plurality of dies **120** via the polymer layer **116**. For example, when the polymer layer **116** first contacts the plurality of dies **120** via the release film **122** an initial force may be applied thereto. Thereafter, as the pressure in the membrane **114** is further increased, the

force may increase, according to $\text{force} = \text{pressure} \times \text{area}$, until the pressure reaches and maintains a target pressure that applies the force F. In some embodiments, the time that passes during application of the initial force and build-up of force up to, but not including, the force F, can be considered part of the second time period T2 described above. In some embodiments, after the force F is reached, the target pressure may be maintained, and the force F may be applied, for a third time period T3 within a range between about 20 minutes and about 30 minutes, such as about 25 minutes.

[0026] As described above, the polymer layer **116** may be formed of a material that is substantially inelastic. The force F from the membrane **114** pushes the polymer layer **116** towards the plurality of dies **120**. Even though the material of the polymer layer **116** is substantially inelastic, the material of the polymer layer **116** is softened at the processing temperature, which is within 50° C. of the Tc of the material of the polymer layer **116**. For example, at the processing temperature, a limited amount of ductility is imparted to the polymer layer **116** such that the polymer layer **116** can take the shape of the gap inside the edge ring **112**. Thus, the portion of the polymer layer **116** located over the plurality of dies **120** may extrude across the gap uniformly. In other words, the inelastic property of the polymer layer **116**, even though softened by the processing temperature, is beneficial to cause the force F to be uniformly applied on the plurality of dies **120**. For example, the bottom surface of the inner radial portion of the polymer layer **116** may be substantially flat when the force F is being applied on the plurality of dies **120**. In some embodiments, the force F may be set based on a processing temperature of the bonding tool **100**. For example, the force F may be positively correlated with processing temperature. Higher processing temperatures (e.g., $T > T_c$) may cause the polymer layer **116** to be more ductile (less brittle) such that higher forces can be applied without causing the polymer layer **116** to crack. Likewise, lower processing temperatures (e.g., $T < T_c$) may cause the polymer layer **116** to be more brittle (less ductile) such that only relatively lower forces can be applied on the plurality of dies **120** to prevent cracking of the polymer layer **116**. For example, when the processing temperature is greater than Tc of the polymer layer **116**, the force F may be greater than 3000 kg, such as within a range between about 3000 kg and about 4000 kg. On the other hand, when the processing temperature is less than Tc of the polymer layer **116**, the force F may be less than about 2000 kg, such as within a range between about 1000 kg and about 2000 kg.

[0027] As described above, the Tc of the material of the polymer layer **116** may not be substantially greater than or less than the processing temperature, such as over or under 50° C. of the processing temperature. If the Tc of the material of the polymer layer **116** is greater than the processing temperature by over 50° C., the polymer layer **116** is too brittle or hard at the processing temperature during the die bonding process. As a result, the polymer layer **116** may crack. On the other hand, if the Tc of the material of the polymer layer **116** is less than the processing temperature by over 50° C., the polymer layer **116** is too ductile or soft at the processing temperature during the die bonding process. As a result, the force F from the membrane **114** may not be applied to the plurality of dies **120** uniformly through the polymer layer **116**. Consequently, some of the dies **120** may not be bonded to the substrate **118**.

[0028] Referring to FIG. 1D and FIG. 2 (block **240**), the membrane **114** is deflated to remove the force F from the plurality of dies **120** and to return the polymer layer **116** to the point of initial contact with the edge ring **112** (e.g., via the release film **122**) with the membrane **114** at the threshold volume (same state as FIG. 1B). As the polymer layer **116** separates from the plurality of dies **120**, the release film **122** enables a clear separation. If the release film **122** is not present, one or more dies **120** may stick to the polymer layer **116** and de-bond from the substrate **118**. In some embodiments, the process at block **240** may occur over a fourth time period T4 within a range between about 60 seconds and about 120 seconds, such as about 90 seconds.

[0029] Referring to FIG. 1E and FIG. 2 (block **250**), the membrane **114** is further deflated to contract the membrane **114**, and the polymer layer **116** attached to the membrane **114**, upwards

away from the substrate **118**. In some embodiments, the further deflation of the membrane **114** at block **250** may coincide with venting pressure from the processing region **108**. As the membrane **114** is deflated, the vertical distance between the bottom surface of the polymer layer **116** and the top surface of the edge ring **112** increases until the membrane **114** returns to the original state (same state as FIG. 1A). Tension in the release film **122** peels the release film **122** away from the polymer layer **116** as the membrane **114** deflates. In some embodiments, the process at block **250** may occur over a fifth time period T5 within a range between about 60 seconds and about 120 seconds, such as about 90 seconds.

[0030] Referring to FIG. 2 (block **260**), the substrate **118** and the plurality of dies **120** that are bonded to the substrate **118** are transferred, via the robot handler, from the substrate support **110** to the factory interface. In some embodiments, the bonding tool **100** is communicatively coupled with and/or includes a controller **124** (FIGS. 1A-1E). The controller **124** may be a workstation computer, a desktop computer, a laptop computer, a tablet computer, or any other type of computing device comprising one or more software products that are capable of controlling processes (e.g., based on input/output of certain operational parameters), receiving process feedback, receiving test result data, performing process adjustments, and/or other software-related control schemes. In some embodiments, the controller **124** includes a processor **126** and a memory **128**. The processor **126** may include a central processing unit (CPU), a graphics processing unit (GPU), a microprocessor, and/or other processing units or components. The memory **128** may be implemented as computer-readable storage media (CRSM), which may be any available physical media accessible by the processor **126** to execute instructions stored on the memory **128**. The memory **128** may be capable of storing various computer readable instructions for performing certain operations described herein (e.g., operations of the controller **124**). The instructions, when executed by the processor **126**, may cause certain operations described herein to be performed. The controller **124** can be communicatively coupled with and control the bonding tool **100** and various equipment (e.g., robot handler, pumps, gas sources, valves, sensors, and other equipment) associated with the bonding tool **100** to implement one or more aspects of the method **200**. For example, the controller **124** can control one or more process timings (e.g., T1, T2, T3, T4, T5), temperatures, pressures, forces, and movements described in connection with FIG. 2. In some embodiments, the controller **124** can be configured to implement all blocks of the method **200** for bonding one or more dies onto a substrate using the bonding tool **100**.

[0031] FIG. 3 is a perspective view of a membrane assembly **300** that includes the membrane **114** with the polymer layer **116** attached, in accordance with some embodiments. The membrane assembly **300** can be attached to the upper tray **104** via a flange **302** that is around an outside portion of the membrane **114**. In some embodiments, the flange **302** may be formed from metal (e.g., stainless steel). The flange **302** may include a plurality of holes **304** that are configured to receive respective fasteners (e.g., screws) that can be used to couple the membrane assembly **300** to the upper tray **104** (FIG. 1A) via respective threaded holes in the upper tray **104**. As shown in FIG. 3, the plurality of holes **304** may be spaced around the circumference of the flange **302**. In some embodiments, the polymer layer **116** may be a circular disc that matches the shape of the edge ring **112** (FIG. 1A). A diameter d1 of the polymer layer **116** may depend on the size of the bonding tool **100** (FIG. 1A) and, in particular, the size of a substrate that is able to fit in the bonding tool **100**. For example, when the substrate is a 300 mm wafer, the diameter d1 of the polymer layer **116** may be within a range between about 310 mm and about 360 mm. In another example, when the substrate is a 200 mm wafer, the diameter d1 of the polymer layer **116** may be within a range between about 210 mm and about 260 mm. In another example, when the substrate is a 450 mm wafer, the diameter d1 of the polymer layer **116** may be within a range between about 460 mm and about 510 mm. As shown in FIG. 3, the polymer layer **116** is radially centered with respect to the membrane **114** and the flange **302**. There is an annular gap **306** between the flange **302** and the polymer layer **116**. In some embodiments, the annular gap **306** may be within a range between

about 20 mm and about 45 mm.

[0032] FIG. 4 is a partial sectional view that illustrates an inflation state **400** of bonding a die **120** onto the substrate **118**, in accordance with some embodiments. As shown in FIG. 4, the inflation state **400** is consistent with the membrane **114** being inflated to the point of initial contact of the polymer layer **116** (via the release film **122**) with the edge ring **112** as described above in connection with FIG. 1B and FIG. 2 (block **220**). As shown in FIG. 4, the diameter $d1$ of the polymer layer **116** is substantially smaller than a diameter $d0$ of the membrane **114** (e.g., when the membrane **114** is deflated). As shown in FIG. 4, the diameter $d1$ of the polymer layer **116** is greater than an inner diameter $d2$ of the edge ring **112** such that an outer portion **402** of the polymer layer **116** overlaps with and rests on an inner portion **404** of the edge ring **112**. In some embodiments, the diameter $d1$ of the polymer layer **116** may be 5% to 20% greater than the inner diameter $d2$ of the edge ring **112**. As shown in FIG. 4, there is a gap $g1$ between a top surface **406** of a die, of the plurality of dies **120**, and the bottom surface of the release film **122**. In some embodiments, the gap $g1$ may range from about 80 μm to about 120 μm , such as from about 90 μm to about 110 μm , such as about 100 μm . The size of the gap $g1$ may be beneficial to cause the force F to be uniformly applied on the plurality of dies **120**. For example, when the gap $g1$ is greater than about 120 μm , the effective force that is applied to one or more dies **120** toward the edge of the substrate **118** may be less than desired (e.g., due to the outer portion **402** of the polymer layer **116** being held in place by resting on the edge ring **112**). On the other hand, in some embodiments, when the gap $g1$ is less than about 80 μm , the effective force that is applied to one or more dies **120** toward the center of the substrate **118** may be greater than desired (e.g., greater than a threshold that can cause one or more dies **120** to crack). As illustrated in FIG. 4, the gap $g1$ is based on a height $h1$ of the edge ring **112** in relation to the combined height $h2$ of the substrate **118** (height, or thickness, $h2s$) and the plurality of dies **120** (height, or thickness, $h2d$). In other words, the gap $g1$ is the difference between the height $h1$ and the combined height $h2$. In some embodiments, a ratio of $h1/h2$ may be within a range between about 1.07 to about 1.11, such as about 1.09. In some embodiments, the height $h1$ of the edge ring **112** may be about 1.2 mm and the combined height $h2$ may be about 1.1 mm (e.g., including the height $h2s$ of the substrate **118** that may be within a range between about 0.725 mm and about 0.775 mm and the height $h2d$ of the plurality of dies **120** that may be within a range between about 0.325 mm and about 0.375 mm). In some embodiments, a ratio of $h2s/h2d$ may be within a range between about 1.9 and about 2.4, such as between about 2.1 and 2.2.

[0033] FIG. 5 is a schematic partial sectional view that illustrates a force application state **500** of bonding a die **120** onto the substrate **118**, in accordance with some embodiments. As shown in FIG. 5, the force application state **500** is consistent with the polymer layer **116** being in contact (e.g., indirectly via the release film **122**) with the plurality of dies **120** and the membrane **114** being inflated to the target pressure that applies the force F , as described above in connection with FIG. 1C and FIG. 2 (block **230**). As shown in FIG. 5, membrane pressure P is applied on a backside of the membrane **114** that faces away from the polymer layer **116**. In some embodiments, a thickness $t1$ of the membrane **114** may be within a range between about 2 mm and about 4 mm, such as about 3 mm. The thickness $t1$ is the actual thickness of the membrane **114** and not the thickness of the inflated membrane **114**. In some embodiments, a thickness $t2$ of the polymer layer **116** may be within a range between about 6 mm and about 10 mm, such as between about 7 mm and about 9 mm, such as about 8 mm. In some embodiments, a thickness $t3$ of the release film **122** may be within a range between about 20 μm and about 80 μm , such as about 50 μm .

[0034] As shown in FIG. 5, the polymer layer **116** may be substantially flat when the force F is being applied on the plurality of dies **120**. When the polymer layer **116** is substantially flat, a bottom surface **408** of the polymer layer **116** is parallel to the top surface **406** of the die **120**. Therefore, the force F is applied primarily to the top surface **406** of the die **120** and in a direction that is substantially perpendicular to a bonding interface between the die **120** and the substrate **118**. This is beneficial to align structures **502** and to prevent shifting of the die **120** with respect to the

substrate **118**, which can cause misalignment of structures **502**, such as conductive features, which is a problem that can occur when conventional bonding tools are used without a polymer layer as disclosed herein. An example of misalignment that can result from non-uniform application of force is shown in phantom.

[0035] FIGS. **6A-6E** illustrate views of various stages of bonding dies onto a substrate, in accordance with some embodiments. FIG. **7** is a flowchart that illustrates an alternative method **700** of bonding dies onto a substrate, in accordance with some embodiments. Aspects of the method **700** that are the same as the method **200** (FIG. **2**) are indicated with reference numerals **210**, **230**, **240** and **260**, and any description in connection therewith is incorporated into the method **700** without limitation. Referring to FIG. **7** (block **210**), a substrate **118** and a plurality of dies **120** that are to be bonded to the substrate **118** are transferred, via a robot handler, from a factory interface to a substrate support **110** of a bonding tool **600**. Referring to FIG. **6A** and FIG. **7** (block **710**), the bonding tool **600** is opened and the polymer layer **116** is manually placed over the edge ring **112**. The polymer layer **116** may be placed on the edge ring **112** with the release film **122** disposed below the polymer layer **116**. The process described in connection with FIGS. **6A-6E** and FIG. **7** may be referred to as a “manual handling” process because the polymer layer **116** is not attached to the membrane **114** and is handled by an operator during the process. Referring to FIG. **6B** and FIG. **7** (block **720**), the membrane **114** of the bonding tool **600** is inflated to expand the membrane **114** downwards towards the substrate **118** and to engage the polymer layer **116**. The process timings described above in connection with block **220** may be incorporated into block **720**, without limitation. Referring to FIG. **6C** and FIG. **7** (block **230**), a force F is applied from the inflating membrane **114** on the plurality of dies **120** via the polymer layer **116**. Referring to FIG. **6D** and FIG. **7** (block **240**), the membrane **114** is deflated to remove the force F from the plurality of dies **120** and to return the polymer layer **116** to the point of initial contact (e.g., via the release film **122**) with the edge ring **112** with the membrane **114** at the threshold volume (same state as FIG. **6B**). Referring to FIG. **6E** and FIG. **7** (block **730**), the membrane **114** is further deflated to contract the membrane **114** upwards away from the substrate **118** and to disengage the polymer layer **116**. The process timings described above in connection with block **250** may be incorporated into block **730**, without limitation. Referring to FIG. **7** (block **740**), the bonding tool **600** is opened and the polymer layer **116** is manually removed from the edge ring **112**, and the used release film **122** is moved away from the processing region **108**. Referring to FIG. **7** (block **260**), the substrate **118** and the plurality of dies **120** that are bonded to the substrate **118** are transferred, via the robot handler, from the substrate support **110** to the factory interface. The manual handling process requires the process to be stopped at two different times per cycle in order to place and remove the polymer layer. Accordingly, the auto handling process (FIG. **2**) can be beneficial in the industry to result in a more efficient process with shorter cycle time and higher throughput compared to the manual handling process (FIG. **7**). In some embodiments, the bonding tool **600** is communicatively coupled with and/or includes a controller **624**. The controller **624** may be a workstation computer, a desktop computer, a laptop computer, a tablet computer, or any other type of computing device comprising one or more software products that are capable of controlling processes (e.g., based on input/output of certain operational parameters), receiving process feedback, receiving test result data, performing process adjustments, and/or other software-related control schemes. In some embodiments, the controller **624** includes a processor **626** and a memory **628**. The processor **626** may include a central processing unit (CPU), a graphics processing unit (GPU), a microprocessor, and/or other processing units or components. The memory **628** may be implemented as computer-readable storage media (CRSM), which may be any available physical media accessible by the processor **626** to execute instructions stored on the memory **628**. The memory **628** may be capable of storing various computer readable instructions for performing certain operations described herein (e.g., operations of the controller **624**). The instructions, when executed by the processor **626**, may cause certain operations described herein to be performed. The controller **624** can be

communicatively coupled with and control the bonding tool **600** and various equipment (e.g., robot handler, pumps, gas sources, valves, sensors, and other equipment) associated with the bonding tool **600** to implement one or more aspects of the method **700**. For example, the controller **624** can control one or more process timings (e.g., **T1**, **T2**, **T3**, **T4**, **T5**), temperatures, pressures, forces, and movements described in connection with FIG. 7. In some embodiments, the controller **624** can be configured to implement all blocks of the method **700** for bonding one or more dies onto a substrate using the bonding tool **600**.

[0036] FIG. 8 is a schematic sectional view that illustrates a first bonding step of forming a semiconductor device package **800**, in accordance with some embodiments. FIG. 9 is a schematic sectional view that illustrates a second bonding step of forming the device structure **800**, in accordance with some embodiments. The first bonding steps and the second bonding step are exemplary illustrations of a process that uses the bonding tool **100** or the bonding tool **600** including the polymer layer **116**, as described herein. Referring to FIG. 8, the semiconductor device package **800** includes a temporary substrate **802** and a first die **801** bonded to the temporary substrate **802**. The first die **801** may include a substrate **806** and an interconnection structure **804**. In some embodiments, the first die **801** may be, or include, a central processing unit (CPU).

[0037] In some embodiments, due to significant mismatch between the coefficients of thermal expansion (CTEs) of the materials of the first die **801**, warpage of the substrate **806** of the first die **801** may occur. For example, as shown in FIG. 8, the warpage of the substrate **806** of the first die **801** can lead to delamination, such that vertical gaps **808** are formed around an outer edge **810** of the first die **801**. The warpage that results from the first bonding step, with the polymer layer **116** as described herein, is reduced compared to a warpage that results from die bonding without the polymer layer **116**. The warpage resulting from die bonding without the polymer layer **116** can create vertical gaps **812** (shown in phantom). In some embodiments, the vertical gap **808** may be within a range between about 11 μm and about 15 μm , while the vertical gap **812** may be within a range between about 14 μm and about 19 μm . In some embodiments, using the polymer layer **116** can reduce warpage by about 20% or more, such as about 20% to about 30%, such as about 30%. As shown in FIG. 8, the delamination as a result of warpage that results from the first bonding step may also include a radial gap **814** around the outer edge **810** of the first die **801**. The radial gap **814** is reduced compared to radial gaps **816** (shown in phantom) that result from die bonding without the polymer layer **116**. In some embodiments, the radial gap **814** may be about 200 μm or less, such as within a range between about 0 μm and about 200 μm , such as between about 0 μm and 100 μm , such as about 0 μm , or between about 3 μm and about 20 μm , such as about 3 μm . In some embodiments, the radial gap **816** may be about 200 μm or more, such as within a range between about 200 μm and about 300 μm , such as about 300 μm .

[0038] As shown in FIG. 9, the second bonding step forms a bond between a second die **901** and the first die **801**. Similar to the first die **801**, the second die **901** includes a substrate **906** and an interconnection structure **904**, and the substrate **906** may have a warpage. The warpage of the substrate **906** can lead to delamination, such that vertical gaps **908** are formed around an outer edge **910** of the second die **901**. The warpage that results from the second bonding step, with the polymer layer **116** as described herein, is reduced compared to a warpage that results from die bonding without the polymer layer **116**. The warpage resulting from die bonding without the polymer layer **116** can create vertical gaps **912** (shown in phantom). In some embodiments, the vertical gap **908** may be within a range between about 4 μm and about 5.5 μm , while the vertical gap **912** may be within a range between about 5 μm and about 7 μm . In some embodiments, using the polymer layer **116** can reduce warpage by about 20% or more, such as about 20% to about 30%, such as about 30%. As shown in FIG. 9, the delamination as a result of warpage that results from the second bonding step may also include a radial gap **914** around the outer edge **910** of the second die **901**. The radial gap **914** is reduced compared to radial gaps **916** (shown in phantom) that results from die bonding without the polymer layer **116**. In some embodiments, the radial gap **914** may be

about 100 μm or less, such as within a range between about 0 μm and about 100 μm , such as between about 0 μm and 50 μm , such as about 0 μm , or between about 3 μm and about 20 μm , such as about 3 μm . In some embodiments, the radial gap **916** may be about 100 μm or more, such as within a range between about 100 μm and about 200 μm , such as about 120 μm . The semiconductor device package **800** may include one or more dummy structures **902** outside the second die **901** and a carrier substrate **907** over the second die **901**. In some embodiments, the second die **901** may be, or include, a static random-access memory (SRAM). After the bonding of the first and second dies **801**, **901**, the temporary substrate **802** and the carrier substrate **907** may be removed, and the resulting package may be a chip-on-wafer (CoW) package.

[0039] Embodiments of the present disclosure provide a bonding tool for bonding dies onto substrates and methods of use. In some embodiments, a method includes positioning a substrate and a plurality of dies over a substrate support of a bonding tool, wherein the plurality of dies are disposed on the substrate, and the substrate is surrounded by an edge ring; inflating a membrane of the bonding tool disposed above the substrate and the edge ring to expand the membrane downwards, wherein a polyetheretherketone (PEEK) layer attached to the membrane rests on the edge ring; and applying a force from the inflating membrane on the plurality of dies via the PEEK layer.

[0040] In some embodiments, a method includes positioning a substrate and a plurality of dies over a substrate support of a bonding tool, wherein the plurality of dies are disposed on the substrate, and the substrate is surrounded by an edge ring; inflating a membrane of the bonding tool disposed above the substrate and the edge ring to expand the membrane downwards, wherein a polymer layer attached to the membrane rests on the edge ring, and the polymer layer has a diameter smaller than a diameter of the membrane when the membrane is deflated; and applying a force from the inflating membrane on the plurality of dies via the polymer layer.

[0041] In some embodiments, a method includes positioning a substrate and a plurality of dies over a substrate support of a bonding tool, wherein the plurality of dies are disposed on the substrate, and the substrate is surrounded by an edge ring; placing a polymer layer on the edge ring, wherein a gap between a top surface of a die of the plurality of dies and a bottom surface of the polymer layer ranges from about 80 μm to about 120 μm ; and inflating a membrane of the bonding tool disposed above the substrate and the edge ring to apply a force on the plurality of dies via the polymer layer.

[0042] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method, comprising: positioning a substrate and a plurality of dies over a substrate support of a bonding tool, wherein the plurality of dies are disposed on the substrate, and the substrate is surrounded by an edge ring; inflating a membrane of the bonding tool disposed above the substrate and the edge ring to expand the membrane downwards, wherein a polyetheretherketone (PEEK) layer attached to the membrane rests on the edge ring; and applying a force from the inflating membrane on the plurality of dies via the PEEK layer.
2. The method of claim 1, wherein the edge ring has a height greater than a combined height of the substrate and a die of the plurality of dies.

3. The method of claim 1, wherein the substrate support is attached to a lower tray of the bonding tool and the membrane is attached to an upper tray of the bonding tool.
 4. The method of claim 3, further comprising controlling respective temperatures of the upper tray and the lower tray, wherein the respective temperatures are configured to be controlled independently.
 5. The method of claim 4, further comprising: maintaining the respective temperatures of the upper tray and the lower tray, relative to a glass transition temperature (T_c) of the PEEK layer, within a range between about -40°C . and about $+30^{\circ}\text{C}$. with respect to the T_c .
 6. The method of claim 1, wherein a processing temperature of the bonding tool is greater than a glass transition temperature (T_c) of the PEEK layer, and wherein the force is within a range between about 3000 kg and about 4000 kg.
 7. The method of claim 1, wherein a processing temperature of the bonding tool is less than a glass transition temperature (T_c) of the PEEK layer, and wherein the force is within a range between about 1000 kg and about 2000 kg.
 8. A method, comprising: positioning a substrate and a plurality of dies over a substrate support of a bonding tool, wherein the plurality of dies are disposed on the substrate, and the substrate is surrounded by an edge ring; inflating a membrane of the bonding tool disposed above the substrate and the edge ring to expand the membrane downwards, wherein a polymer layer attached to the membrane rests on the edge ring, and the polymer layer has a diameter smaller than a diameter of the membrane when the membrane is deflated; and applying a force from the inflating membrane on the plurality of dies via the polymer layer.
 9. The method of claim 8, wherein the polymer layer has an elasticity smaller than an elasticity of the membrane.
 10. The method of claim 8, wherein inflating the membrane further comprises causing a bottom surface of the polymer layer to be in contact with a release film that is positioned between the polymer layer and the plurality of dies.
 11. The method of claim 8, wherein the edge ring has a height substantially greater than a combined height of the substrate and a die of the plurality of dies.
 12. The method of claim 11, wherein the height of the edge ring is greater than the combined height of the substrate and the die by about $80\text{ }\mu\text{m}$ to about $120\text{ }\mu\text{m}$.
 13. The method of claim 8, wherein the polymer layer comprises polyetheretherketone (PEEK).
 14. The method of claim 8, wherein the diameter of the polymer layer is 5% to 20% greater than an inner diameter of the edge ring.
 15. A method, comprising: positioning a substrate and a plurality of dies over a substrate support of a bonding tool, wherein the plurality of dies are disposed on the substrate, and the substrate is surrounded by an edge ring; placing a polymer layer and a release film on the edge ring, wherein a gap between a top surface of a die of the plurality of dies and a bottom surface of the release film ranges from about $80\text{ }\mu\text{m}$ to about $120\text{ }\mu\text{m}$; and inflating a membrane of the bonding tool disposed above the substrate and the edge ring to apply a force on the plurality of dies via the polymer layer.
 16. The method of claim 15, wherein the polymer layer comprises polyetheretherketone (PEEK).
 17. The method of claim 15, wherein a diameter of the polymer layer is 5% to 20% greater than an inner diameter of the edge ring.
 18. The method of claim 15, wherein the membrane comprises silicone.
 19. The method of claim 15, wherein inflating the membrane further comprises: causing an inner radial portion of the polymer layer to advance across the gap to rest on the plurality of dies.
 20. The method of claim 19, wherein a bottom surface of the inner radial portion of the polymer layer is substantially flat when the force is being applied on the plurality of dies.
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